

Volume 25

Sentinel Mission Operating System (SMOS)

Version: 1.0

Status: Draft

Vision

Everything a user does inside Project Sentinel is a Mission.

Not:

- Scan
- Tool
- Module
- Wizard
- Script

But a Mission.

Examples:

- Secure My Website
- Check My Computer
- Analyze My Mobile Phone
- Audit Cloud Environment
- Review Identity Security
- Prepare for ISO 27001
- Perform an Authorized Bug Bounty Assessment
- Investigate Suspicious Activity
- Verify Security Improvements

This creates one consistent operational model.

Chapter 1

Mission Object

Every mission is an object.

Example

Mission ID

Chapter 2

Mission Lifecycle

Created

Every mission follows this lifecycle.

Chapter 3

Mission Planner

Instead of the user selecting tools, they define an objective.

Example

Secure my website.

The Mission Planner automatically builds the execution plan.

It selects:

- Discovery Engine.
- HTTP Engine.
- TLS Engine.
- Header Engine.
- Technology Engine.
- Accessibility Engine.
- Reporting Engine.

The user focuses on the goal, not the implementation.

Chapter 4

Mission Profiles

Profiles define depth and scope.

Examples:

Quick

Designed for rapid health checks.

Standard

Balanced assessment for routine operations.

Deep

Comprehensive assessment of the authorized environment.

Compliance

Evidence collection mapped to selected frameworks.

Continuous

Recurring monitoring with scheduled reassessment.

Profiles influence orchestration rather than individual tools.

Chapter 5

Mission Policies

Organizations can define:

- Approved scanning windows.
- Maximum execution time.
- Required approvals.
- Notification preferences.
- Evidence retention.
- Reporting requirements.

Policies are inherited by missions.

Chapter 6

Mission Evidence

Every mission builds an evidence package.

Contains:

- Evidence objects.

- Findings.
- Trust changes.
- Screenshots.
- Configuration observations.
- Verification results.
- Timeline.

Nothing is separated.

Chapter 7

Mission Timeline

Example

08:00

Chapter 8

Mission Replay

Every mission can be replayed.

The user can inspect:

- Decisions.
- Evidence.
- Engine outputs.
- Timing.
- Relationships.
- Generated recommendations.

This improves transparency and troubleshooting.

Chapter 9

Mission Comparison

Organizations compare missions.

Example

January

↓

April

↓

July

↓

October

The platform highlights:

- Improvements.
 - New risks.
 - Resolved findings.
 - Technology changes.
 - Trust evolution.
-

Chapter 10

Mission Library

Organizations build reusable mission templates.

Examples

Weekly Website Review

Monthly Endpoint Review

Daily Mobile Health Check

Cloud Baseline Assessment

Supplier Security Review

Executive Dashboard Refresh

These become reusable operational building blocks.

PS-ADR-0022

Mission-First Architecture

Status

Accepted

Decision

Users interact with Project Sentinel through missions rather than individual engines.

Rationale

Users think in objectives, while the platform manages technical orchestration.

NEW CORE ENGINE

Mission Optimization Engine

After every mission, the platform evaluates:

- Duration.
- Resource usage.
- Evidence quality.
- Rule effectiveness.
- Duplicate work.
- User feedback.

Future missions become more efficient.

NEW CORE ENGINE

Mission Quality Engine

Every mission receives a quality score.

Based on:

Coverage

Evidence completeness

Verification rate

Confidence

Policy compliance

Execution success

This measures the quality of the assessment rather than simply the number of findings.

NEW CORE ENGINE

Mission Cost Engine

Particularly valuable for enterprises.

Tracks:

- CPU usage.
- Memory usage.
- Storage consumption.
- Network traffic.
- Execution time.

- Cloud costs (if applicable).

Organizations understand the operational cost of security activities.

NEW IDEA

Mission Marketplace

Organizations can share or distribute mission templates internally.

Examples:

- Secure WordPress Deployment.
- New Kubernetes Cluster Review.
- Employee Laptop Baseline.
- Retail Store Audit.
- Financial Services Baseline.

Templates remain version-controlled and can be digitally signed.

NEW IDEA

Mission Coach

Instead of only executing tasks, Project Sentinel explains what is happening.

For example:

"This mission is now verifying TLS configuration because encrypted transport is essential for protecting communications."

Or:

"The Accessibility Engine is checking keyboard navigation to support WCAG 2.2 AA compliance."

This turns every mission into a learning opportunity.

NEW IDEA

Adaptive Mission Engine

Future versions may allow missions to adapt automatically based on observed evidence.

Example:

1. Discovery identifies an exposed API.
2. The API is confirmed to be within the authorized scope.
3. The Mission Planner adds the API Assessment Engine.
4. New evidence updates the execution plan.
5. The final report reflects the expanded assessment.

Every adaptation is recorded, explained, and reviewable.

PROJECT SENTINEL ROADMAP — Next Engineering

Phase

After defining the Mission Operating System, the platform is ready to enter subsystem-level engineering.

The next volumes will specify:

Volume 26 — Endpoint Security Engine

- Windows
- Linux
- macOS
- Endpoint Health
- Persistence Detection
- Configuration Assessment
- Evidence Collection

Volume 27 — Mobile Security Engine

- Android
- iOS
- Device Health
- Application Inventory
- Permission Analysis
- Configuration Review
- Security Guidance

Volume 28 — Identity Security Engine

- Local identities
- Directory integrations
- MFA posture
- Privilege analysis
- Service accounts
- Trust relationships

Volume 29 — Cloud Security Engine

- AWS
- Azure
- Google Cloud
- Containers
- Kubernetes
- Identity
- Storage
- Network policies

Volume 30 — Security Operations & Response

- SOC workspace
- Incident management
- Case management
- Evidence-driven investigations
- Detection rules

- MITRE ATT&CK mapping
 - Workflow automation
-

One New Long-Term Vision

I would like to introduce one final concept that I believe can become the signature capability of Project Sentinel.

The Sentinel Cyber Genome™

Every managed object—website, server, laptop, mobile device, cloud resource, identity, or application—receives a **Cyber Genome**.

The Cyber Genome is not marketing terminology; it is a structured profile composed of immutable identity information, observed characteristics, security posture, trust history, relationships, lifecycle events, compliance mappings, and evidence-backed behavioral baselines.

Rather than asking:

"What vulnerabilities does this server have today?"

Project Sentinel can answer:

- How has this server evolved over three years?
- Which security controls improved its trust?
- Which recurring configuration changes introduce risk?
- Which identities interact with it most frequently?
- Which applications depend on it?
- Which verified remediations have historically reduced risk?

The Cyber Genome transforms isolated assessments into a continuously evolving understanding of each object's security journey.

A note on implementation discipline

As we continue, we should maintain one important engineering rule:

Every blueprint volume must correspond to a future Python package or service.

That traceability ensures our documentation, architecture, and implementation evolve together. It also means that when we eventually start coding, we will not be inventing the architecture—we will simply be implementing a design that has already been carefully reviewed and refined. This approach keeps the project ambitious while remaining grounded in practical, maintainable engineering.

====[USER]====

Good and continue

====[ASSISTANT]====

Excellent. We are now entering what I believe will become the **largest engineering part** of Project Sentinel.

Everything we have designed so far (Object Engine, Mission Engine, Knowledge Engine, Evidence Engine, Digital Twin, Trust Engine) now becomes the foundation for what I call:

Sentinel Endpoint Intelligence Platform (SEIP)

I intentionally do **not** call it an Antivirus or EDR.
Those are only small parts of what we are building.
Our goal is much larger.

PROJECT SENTINEL BLUEPRINT

Volume 26